

IoT-Based Soldier Security, Health Monitoring & Emergency Rescue System

Developer: THIRUPATHI AASHRITH

Affiliation: Undergraduate Student, Department of Electronics and Communication Engineering (ECE)

GitHub Repository: <https://github.com/thirupathiaashrith6831>

1. Project Overview & Objective

Military personnel operate in remote, high-altitude, and hostile environments where extreme weather, toxic hazards, and trauma pose severe operational risks. The objective of this project is to build an intelligent, multi-sensor wearable telemetry unit using the **ESP32** microcontroller. The system performs continuous automated data logging to **ThingSpeak** across six distinct parameters and provides a dedicated distress trigger that activates a local acoustic/visual alarm while transmitting real-time satellite coordinates and biometric data to commanders via Telegram.

2. Soldier's Perspective: Tactical Significance

- **Remote Tactical Triage:** Command stations track soldier vitals (heart rate simulation) alongside environmental conditions to detect fatigue, hypothermia, or heatstroke remotely.
- **Barometric & High-Altitude Monitoring:** With the onboard BMP180 sensor, base command continuously tracks atmospheric pressure and altitude, critical for high-altitude missions and airborne operations.
- **Toxic Gas & Smoke Warning:** The MQ sensor continuously samples ambient air quality, alerting against chemical hazards or smoke inhalation.
- **One-Touch SOS Extraction:** In the event of injury, ambush, or separation, pressing the tactile panic button immediately sounds a 1000 Hz distress beacon, flashes a high-visibility Red LED, and dispatches a Google Maps extraction link directly to the field operations Telegram channel.
- **Silent Operational Tracking:** Normal operation is indicated locally by a pulsing Green LED heartbeat, while telemetry uploads automatically every 20 seconds without requiring manual radio reports.

3. Hardware Requirements

- **ESP32 DevKit V1 (30-pin):** Central Wi-Fi/BLE microcontroller.
- **u-blox NEO-6M GPS Module:** Global satellite positioning receiver with external patch antenna.
- **DHT11 Sensor Module:** Digital sensor for ambient temperature and relative humidity.
- **BMP180 Barometric Pressure & Altitude Sensor:** I2C sensor measuring barometric pressure (hPa) and altitude (m).

- **MQ-Series Gas Sensor (Flying-Fish PCB):** Analog sensor for air quality and hazardous gas detection.
- **Active/Passive Buzzer:** Acoustic signaling device driven at 1000 Hz during emergency distress.
- **Green & Red 5mm LEDs:** Visual status indicators (Green = System Normal, Red = SOS Active).
- **Current Limiting Resistors (220Ω – 330Ω):** For LED protection.
- **Tactile Push Button:** Active-low emergency SOS input.
- **Breadboard & Jumper Wires:** For circuit interconnection.
- **Power Source:** 5V portable power bank or 3.7V Li-ion battery with boost regulator.

4. Hardware Pin Mapping & Circuit Layout

All components share a unified system **GND** connected directly to the ESP32 ground rail.

Component	Pin on Component	ESP32 Pin	Interface Type	Functional Role
DHT11 Module	DATA / OUT	GPIO 4	Digital I/O	Temperature & humidity measurement
BMP180 Sensor	SDA	GPIO 21	I2C (Wire)	Serial data line
BMP180 Sensor	SCL	GPIO 22	I2C (Wire)	Serial clock line
NEO-6M GPS Module	TX	GPIO 16 (RX2)	Hardware UART 2	NMEA sentence serial transmission
NEO-6M GPS Module	RX	GPIO 17 (TX2)	Hardware UART 2	Configuration command receiver
Tactile Push Button	Terminal 1	GPIO 12	Digital Input	SOS trigger (INPUT_PULLUP)
Tactile Push Button	Terminal 2	GND	Ground	Pulls GPIO 12 LOW when

				pressed
Buzzer	Positive (+)	GPIO 13	Digital Output (PWM)	1000 Hz distress tone generator
Buzzer	Negative (-)	GND	Ground	Ground reference
Green Status LED	Anode (+)	GPIO 14	Digital Output	1 Hz heartbeat blink
Green Status LED	Cathode (-) via 220Ω	GND	Ground	Current limit to ground
Red Alert LED	Anode (+)	GPIO 27	Digital Output	SOS strobe warning flasher
Red Alert LED	Cathode (-) via 220Ω	GND	Ground	Current limit to ground
MQ Gas Sensor	AO (Analog Out)	GPIO 34	ADC1 (Input Only)	Ambient air quality voltage reading
Floating Reference	Open Pin	GPIO 35	ADC1 (Input Only)	randomSeed() true random entropy
Power Distribution	VCC / VIN	3V3 / VIN (5V)	Power Bus	3.3V (BMP180, DHT11, GPS) & 5V (MQ Sensor)
Ground Distribution	GND	GND	Power Bus	Unified common ground plane

5. Cloud Platform & Telemetry Configuration

A. ThingSpeak Setup (Multi-Field Dashboard)

1. Log in to [ThingSpeak](#) and navigate to **Channels > My Channels > New Channel**.
2. Set the Channel Name to Soldier Security & Health Monitoring.
3. Configure the 6 telemetry fields:
 - **Field 1:** Temperature (°C)
 - **Field 2:** Humidity (%)
 - **Field 3:** Air Quality / Gas
 - **Field 4:** Heart Rate (BPM)
 - **Field 5:** Pressure (hPa)
 - **Field 6:** Altitude (m)
4. Save the channel, click on the **API Keys** tab, and copy your **Write API Key**.
5. Note your numeric **Channel ID** (3272768) from the channel dashboard.

B. Telegram Bot Setup (Emergency Extraction Protocol)

1. Open Telegram and search for @BotFather.
2. Issue /newbot, provide a name, and choose a unique username ending in bot.
3. Copy the **HTTP API Bot Token** provided.
4. Search for @myidbot and send /getid to retrieve your unique numerical **Chat ID** (6203499767).
5. Open your bot's chat window and press /start to allow automated message delivery.

6. System Working Methodology

1. **Initialization:** On boot, the ESP32 starts standard UART2 at 9600 baud for GPS streaming, initializes I2C on GPIO 21/22 for the BMP180 barometric sensor, configures internal pull-ups on GPIO 12, initializes the random seed via analog noise on GPIO 35, and connects to the configured Wi-Fi hotspot.
2. **Heartbeat & Telemetry Acquisition:**
 - The **Green LED** toggles every 1 second without blocking code execution using millis() timing.
 - Ambient temperature and relative humidity are sampled from the **DHT11**.
 - Barometric atmospheric pressure and computed altitude are read from the **BMP180**.
 - Air quality index is captured from the **MQ sensor** analog line.
 - Heart rate values are indexed dynamically to represent physiological load.
3. **Cloud Logging Routine:** Every 20 seconds, the 6-parameter payload is compiled and transmitted using ThingSpeak.writeFields() to update the real-time cloud analytics charts.
4. **Distress Sequence (Button Press):**
 - The Green LED turns off immediately.
 - The **Buzzer** emits a 1000 Hz emergency tone while the **Red LED** strobes for 3 seconds.
 - The ESP32 parses the latest valid latitude and longitude from the NEO-6M NMEA stream.

- An emergency HTTPS payload containing all telemetry values and a clickable Google Maps location link
(`[https://www.google.com/maps?q=lat,lng](https://www.google.com/maps?q=lat,lng)`) is delivered to the command channel via the Telegram Bot API.

7. Conclusion

The **IoT-Based Soldier Security, Health Monitoring & Emergency Rescue System** provides a comprehensive, field-ready framework for situational awareness and tactical safety. By unifying barometric altimetry, environmental sensing, physiological modeling, satellite navigation, and dual-channel reporting (ThingSpeak and Telegram), the system ensures rapid extraction response and robust health surveillance for military personnel operating in critical operational zones.